

Mock Interview Platform for Interview Performance Analysis using Machine Learning

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Abstract—In today's competitive job market, excelling in technical interviews is vital for students transitioning from college to careers. Traditional preparation methods often fail to address individual needs, career goals, and specific domains. This paper introduces an Automated Interview System leveraging advanced Natural Language Processing (NLP) and deep learning to provide a personalised and realistic interview experience. Users can select their desired domain, and the system dynamically generates interview questions in real-time using a custom dataset, eliminating dependence on external tools like Google Gemini. The system offers performance evaluation and detailed feedback on improvement areas. Additionally, the paper discusses the interface design, emphasising a seamless user experience. This research showcases the system's potential to enhance interview preparation for college students.

Keywords— *Natural Language Processing (NLP), Deep Learning, Automated Interview System, Dynamic Question Generation, Real-time Adaptation, Candidate Performance Evaluation.*

I. INTRODUCTION

In today's competitive job market, recent graduates face challenges in preparing effectively for technical interviews. Traditional methods like studying question banks or mock interviews often lack personalisation, leaving candidates unclear about their strengths and areas for improvement. Advances in Artificial Intelligence (AI) and Machine Learning (ML) offer transformative solutions. AI-powered interview platforms have demonstrated significant improvements in providing personalized feedback and dynamic question generation. Patil et al. [1] highlighted the use of AI-driven mock interviews to improve performance through adaptive question generation. Purohit [2] developed an NLP-based chatbot that evaluates candidate responses and provides instant feedback. Mohri et al. [3] introduced an AI-based dynamic model for generating question papers in adaptive assessments, enhancing personalization and adaptability. Dharmatti et al. [4] proposed a smart interview system integrating voice, text, and emotion analysis for a more comprehensive evaluation. Massey et al. [5] explored the application of machine learning in traceability links classification, providing insights into automated assessment methods. This project introduces an Automated Interview System designed for college students transitioning to the

workforce. Using advanced Natural Language Processing (NLP) and deep learning, the system delivers a personalised, realistic interview experience. Instead of relying on resumes, users select a specific domain or skill area for their interview. The system dynamically generates and adapts questions in real-time based on the candidate's responses, creating an interactive and adaptive simulation. This innovative approach addresses the limitations of traditional methods, offering targeted preparation and actionable feedback for career success.

II. RELATED WORK

Research on AI-powered interview technologies explores several innovative techniques to enhance the hiring process. AI chatbots [2] integrate NLP, Naïve Bayes, Phrase Reinforcement, and Sentiment Analysis to optimize interviews. Mohri et al. [3] introduced a dynamic model to generate question papers in adaptive assessments, improving personalization. Dharmatti et al. [4] proposed a smart interview system that integrates voice, text, and emotion analysis to enhance candidate evaluation. Additionally, Massey et al. [5] explored machine learning-based traceability links classification, offering a new dimension to automated assessment methods. These technologies enable features like resume analysis, automatic question generation, and real-time sentiment evaluation, allowing for personalized, adaptive interviews.

[6] introduced a machine learning-based traceability links classification system, enhancing automated assessment techniques. Sree et al. [7] proposed a multilinear subspace learning method for dimensionality reduction, improving performance in handling high-dimensional data. Brownlee [8] explored the transformative impact of deep learning on technology, showcasing advancements in image recognition and natural language processing. Zheng et al. [9] developed a natural language processing model for generating questions from keywords with user refinements, enhancing the adaptability of interview systems. Robnik-Šikonja and Kononenko [10] focused on feature selection methods using the Relief algorithm, improving model accuracy through effective feature extraction.

Optimization algorithms and learning methods have advanced considerably over the last few years, which has played an important role in the performance and flexibility of deep learning models across various fields. Kingma and Ba proposed the Adam optimizer as one of the most widely used stochastic optimization techniques owing to its adaptive learning rates and efficient convergence [11]. This optimiser has been used to train deep architectures like the Deep Convolutional Neural Network (CNN) proposed by Krizhevsky et al. for ImageNet classification, a landmark event in computer vision research [12]. Recent research by Aytekin has advanced a new perspective of neural networks as decision trees, providing theoretical explanation of model interpretability and structure [13].

In real-world applications, especially within the field of human-computer interaction and affective computing, deep learning has made it possible to automate subjective evaluations. Nguyen et al. designed an automated interview assessment system based on deep learning models that performed robust classification of candidate answers on the basis of linguistic and visual indications [14]. Likewise, Su et al. introduced a framework for forecasting behavioral competencies from real-time video interviews using facial expressions, demonstrating the intersection of image processing and psychological evaluation methods [15].

TABLE I
 A SUMMARY OF RESEARCH REVIEWED IN SECTION II

Paper Reference	ML Algorithms	Features	Data Traces
Patil et al. [1] -2023	CNNs, NLP, Random Forests, SVMs, and LSTMs for mock interviews.	Focuses on emotion analysis, confidence assessment	FER-2013, AffectNet, and NLP tools like NLTK and spaCy.
Purohit [2] - 2021	Naïve Bayes Algorithm, Phrase Reinforcement Algorithms, and Sentiment	Resume analysis, automatic question generation, sentiment analysis.	Resume data, candidate responses, Google Firebase, and dynamic question
Mohri et al. [3] — 2018	Dynamic Model, Adaptive Algorithms	Dynamic model for generating question papers in adaptive assessments	Educational Data Mining datasets
Dharmatti et al. [4] — 2022	LSTM, RNN, Sentiment Analysis	Automates the interview process with NLP	Interview datasets and response logs.
Massey et al. [5] — 2023	Classification Algorithms	Traceability link classification for software engineering	Software repositories datasets
Lu et al. [6] — 2013	Multilinear Subspace Learning	Dimensionality reduction for multidimensional data	Image and video datasets
Sree [7] — 2025	Deep Learning	Revolutionizing technology through deep learning	Applications in image recognition and NLP using large datasets
Zheng et al. [8] — 2013	Natural Language Processing (NLP)	Natural Language Processing (NLP)	Natural Language Processing (NLP)

Robnik-Šikonja et al. [9] — 2003	Relief Algorithm	Feature selection in machine learning	Tested on various datasets for feature selection
Vaswani et al. [10] — 2017	Attention Mechanism	Transformer model for processing sequential data	Machine translation datasets
Kingma et al. [11] — 2015	Adam Optimizer	Adaptive learning rate optimization algorithm	Tested on multiple optimization tasks
Krizhevsky et al. [12] — 2012	Convolutional Neural Networks (CNN)	Image classification with deep CNNs	ImageNet dataset
Aytekin et al. [13] — 2019	Neural Networks, Decision Trees	Interpreting neural networks as decision trees	Various datasets for model interpretation

III. MACHINE LEARNING ALGORITHMS

A. Reinforcement Learning Algorithm for Dynamic Question Generation

Reinforcement Learning (RL) is used to dynamically generate interview questions based on candidate performance. The algorithm adjusts question difficulty in real-time to ensure an adaptive interview experience.

Algorithm Steps:

1. Initialise Q-values: Define a state-action space where states represent the user's current performance level, and actions correspond to selecting a question of a certain difficulty.
2. Choose an initial question randomly from the question bank.
3. Observe candidate response and assign a reward: If the answer is correct and confident → assign a higher reward, If the answer is incorrect or lacks confidence → assign a lower reward
4. Update Q-value using the Q-learning formula:
$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma a' \max_{a'} Q(s',a') - Q(s,a)]$$
5. Adjust difficulty dynamically based on the updated Q-values.
6. Repeat until all interview questions are answered or time runs out.
7. End.

B. Natural Language Processing (NLP) for Response Evaluation

NLP techniques evaluate the candidate's answer by analyzing semantic similarity with the expected answer.

Algorithm Steps:

1. Convert the candidate's response and the expected answer into vector representations using word embeddings (e.g., Word2Vec, TF-IDF, or BERT).
2. Compute the Cosine Similarity between the two vectors:

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

A = vector representation of the candidate's response

B = vector representation of the expected answer

3. If similarity score >>> threshold → mark answer as correct; otherwise, mark as incorrect or partially correct.
4. Apply additional NLP techniques such as sentiment analysis (to assess confidence) and syntactic parsing (to check grammar correctness).
5. Store evaluation results for further analysis.
6. End.

C. Hashing Algorithm for Secure Authentication

To ensure user security, passwords are stored using a cryptographic hashing function.

Algorithm Steps:

1. User enters a password during login.
2. Compute the hashed value of the entered password:

$$H(\text{password}) = \text{Hash Function}(\text{input})$$

3. Compare the computed hash with the stored hash in the database.
4. If hashes match → authentication is successful; otherwise → access is denied.
5. End.

D. Adaptive Feedback Scoring using Weighted Sum Model

The system provides real-time feedback based on correctness, clarity, and confidence.

Algorithm Steps:

1. Assign weight values w_1, w_2, w_3 to different aspects of the response:
 C = correctness score, C_1 = clarity score, C_o = confidence score
2. Compute the final feedback score as:

$$\text{Feedback Score} = w_1 \times C + w_2 \times C_1 + w_3 \times C_o$$

3. Display feedback to the candidate.
4. Adjust weights dynamically based on system learning.
5. End.

E. Difficulty Adjustment using Performance-Based Scaling

The next question's difficulty is adjusted based on the candidate's performance.

Algorithm Steps:

1. Define performance levels and target thresholds.
2. Compute the performance difference:
 Δ = performance – target level
3. Adjust difficulty using:

$$D_{next} = D_{current} + \beta \times \Delta, \text{ where } \beta \text{ is the adjustment rate.}$$

4. If candidate performance is high $\rightarrow$ increase question difficulty.
5. If performance is low $\rightarrow$ reduce difficulty to reinforce foundational knowledge.
6. Repeat until interview completion.
7. End.

IV. PROJECT IMPLEMENTATION AND TESTBED FOR EXPERIMENTATION

The Fig. 1. Interview Module with Dynamic Question Generation, shows the workflow of a mock interview platform. It starts by selecting a specific interview domain and using web scraping to collect relevant domain questions. The candidate's audio responses are converted to text and compared with expected answers using NLP. If the similarity score surpasses 50%, the next question is generated based on the response. The system adjusts question complexity dynamically and stores responses for further evaluation, offering personalised feedback to enhance both technical and soft skills. The platform simulates real-time interviews, refining questions based on performance to provide a tailored and evolving interview experience.

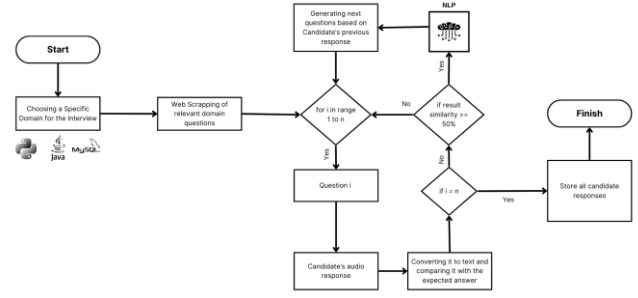


Fig. 1. Interview Module with Dynamic Question Generation

Perform Evaluation Parameters

1. Response Accuracy (RA): Measures the percentage of correct answers provided by the user. It can be expressed as:

$$RA = \frac{\text{Number of Correct Answers}}{\text{Total Number of Questions}}$$

2. Question Difficulty Adaptation (QDA): This evaluates how well the system adapts the difficulty of the questions based on user performance. It is computed based on the difficulty level increase or decrease relative to user performance trends.

$$QDA = \frac{\text{Number of Adjusted Questions}}{\text{Total Questions Asked}}$$

3. Feedback Effectiveness (FE): Measures the impact of real-time feedback on user performance improvement. It calculates how much the user's accuracy improves after receiving feedback.

$$FE = \frac{\text{Post Feedback Correct Answers} - \text{Pre Feedback Correct Answers}}{\text{Total Number of Questions}}$$

4. Response Time Efficiency (RTE): Evaluates how efficiently users are able to answer the questions within a given time. It calculates the average response time per question.

$$RTE = \frac{\text{Total Response Time}}{\text{Total Number of Questions}}$$

5. Interview Completion Rate (ICR): Measures the proportion of users who complete the interview session versus those who abandon it mid-way.

$$ICR = \frac{\text{Number of Users who completed}}{\text{Total Users who started}}$$

V. RESULTS AND DISCUSSIONS

A. Methodology: The platform integrates Machine Learning (ML) and Natural Language Processing (NLP) for dynamic question generation and Reinforcement Learning for adaptive difficulty adjustment. For candidate evaluation, a Convolutional Neural Network (CNN)-based Face Emotion Detection Model is employed to analyze facial expressions during interviews. The model processes 48x48 grayscale facial images through multiple convolutional layers with increasing filters (128, 256, 512) and ReLU activation, followed by max-pooling layers for down-sampling and dropout layers to prevent overfitting. The extracted

features are passed through fully connected dense layers before producing a probability distribution over seven emotion classes using a SoftMax activation function. The Adam optimizer with a learning rate of 0.001 is used for training, with categorical cross-entropy as the loss function. Real-time emotion predictions provide insights into the candidate's emotional state, helping assess confidence levels and engagement. Additionally, cosine similarity measures response accuracy, and a weighted feedback system evaluates correctness, clarity, and emotional stability.

B. Result & Analysis: The Face Emotion Detection Model enhanced the platform's ability to assess candidate emotions and confidence during interviews. The CNN-based model achieved an accuracy between 70% and 75%, effectively distinguishing emotions like *Neutral*, *Happy*, *Sad*, and *Surprised*, enriching candidate evaluation beyond textual or verbal responses alone. Real-time emotion tracking enabled the platform to dynamically adapt interview difficulty, offering a personalized experience. While the model performed well overall, minor misclassifications occurred, particularly between emotions like *Surprise* and *Fear*, especially in low-light conditions or with ambiguous expressions. Integrating emotion detection provided deeper insights into non-verbal cues, improving performance reports and refining the assessment of candidates' emotional states throughout the interview.

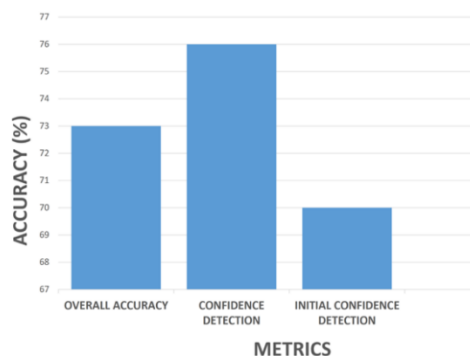


Fig 2. Metric VS Accuracy

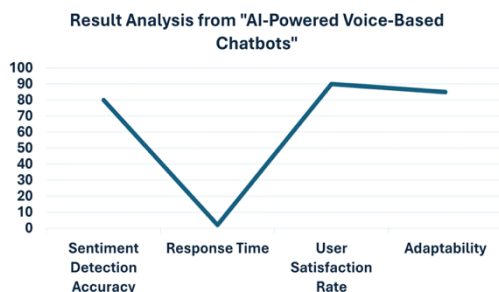


Fig. 3. Method VS Accuracy (AI-Powered Voice-Based Chatbots)

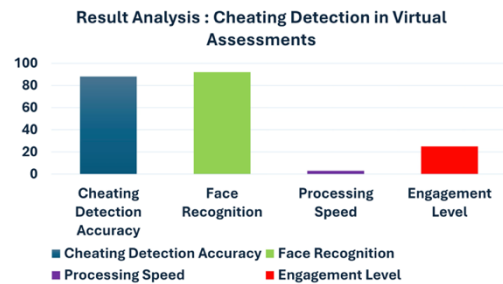


Fig. 4. Metric VS Value (Cheating detection in Virtual Assessments)

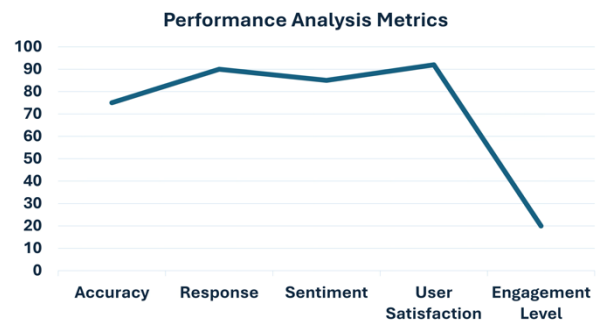


Fig. 5. Metric VS Value

The figures collectively illustrate the performance and effectiveness of various AI-driven techniques used in the mock interview platform. Fig. 2 showcases the accuracy of the Face Emotion Detection Model, highlighting its role in enhancing confidence detection by integrating facial expression analysis. The model achieved an overall accuracy between 70% and 75%, effectively identifying emotions that provided deeper insights into candidates' emotional states during interviews. Fig. 3 highlights the effectiveness of different evaluation methods, revealing that combining facial expressions with audio sentiment analysis significantly enhances accuracy in assessing candidate confidence and engagement. Fig. 4 focuses on cheating detection mechanisms in virtual assessments, comparing traditional methods with AI-based approaches like face recognition and behavioral pattern analysis, demonstrating improved detection accuracy. Fig. 5 provides an overall performance assessment of the system across various metrics, emphasizing how adaptive difficulty scaling and real-time feedback enhance user experience and skill development over multiple interview sessions. These findings reinforce the importance of integrating AI-driven methodologies to create a more personalized and effective interview simulation platform.

VI. CONCLUSION AND FUTURE SCOPE

The Mock Interview Platform's future development will focus on enhancing candidate evaluation through emotional intelligence, sentiment analysis, and VR-based simulations to create realistic interview environments. Integrating advanced facial emotion detection with audio sentiment analysis will provide deeper insights into

candidates' non-verbal cues, enabling a more holistic assessment. Multilingual support will improve accessibility, allowing candidates from diverse linguistic backgrounds to benefit from personalized mock interviews. Additionally, AI-driven personalized learning will refine preparation strategies by adapting the difficulty of questions based on performance, ensuring continuous improvement. Blockchain integration can ensure secure data storage, enhancing the reliability and transparency of candidate records while protecting sensitive interview data. Incorporating real-time feedback loops will empower candidates to address areas of improvement instantly, fostering a more interactive experience. The platform could also explore integrating biometric verification and gaze tracking for enhanced proctoring, reducing the likelihood of fraudulent activities during assessments.

Furthermore, partnerships with hiring platforms and academic institutions will expand its impact by bridging the gap between education and employment. Collaborations with industry experts can facilitate the creation of domain-specific question banks, ensuring that candidates receive relevant training aligned with market demands. Future research can focus on refining the reinforcement learning algorithms to make the question generation process even more adaptive and tailored to individual candidate needs.

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